Dietary fiber intake and risk of first stroke: a systematic review and meta-analysis.

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BACKGROUND AND PURPOSE: Fiber intake is associated with reduced stroke risk in prospective studies, but no meta-analysis has been published to date. METHODS: Multiple electronic databases were searched for healthy participant studies reporting fiber intake and incidence of first hemorrhagic or ischemic stroke, published between January 1990 and May 2012. RESULTS: Eight cohort studies from the United States, northern Europe, Australia, and Japan met inclusion criteria. Total dietary fiber intake was inversely associated with risk of hemorrhagic plus ischemic stroke, with some evidence of heterogeneity between studies (I(2); relative risk per 7 g/day, 0.93; 95% confidence interval, 0.88-0.98; I(2)=59%). Soluble fiber intake, per 4 g/day, was not associated with stroke risk reduction with evidence of low heterogeneity between studies, relative risk 0.94 (95% confidence interval, 0.88-1.01; I(2)=21%). There were few studies reporting stroke risk in relation to insoluble fiber or fiber from cereals, fruit, or vegetables. CONCLUSIONS: Greater dietary fiber intake is significantly associated with lower risk of first stroke. Overall, findings support dietary recommendations to increase intake of total dietary fiber. However, a paucity of data on fiber from different foods precludes conclusions regarding the association between fiber type and stroke. There is a need for future studies to focus on fiber type and to examine risk for ischemic and hemorrhagic strokes separately.